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Conducting a Security Audit and Risk Assessment

1. Aim

To perform a security audit on a Windows system and identify potential vulnerabilities by conducting a risk assessment and suggesting mitigation strategies.

2. Tools Required

- Windows Operating System (Windows 10/11)
- Windows Security (Defender / Antivirus)
- Command Prompt
- Task Manager
- Web Browser (Google Chrome / Microsoft Edge)

3. Introduction

A security audit is the process of evaluating a system's security measures to identify vulnerabilities. Risk assessment involves analyzing potential threats, vulnerabilities, and their impact on the system. It helps in implementing appropriate security controls to protect data and resources from cyber threats.

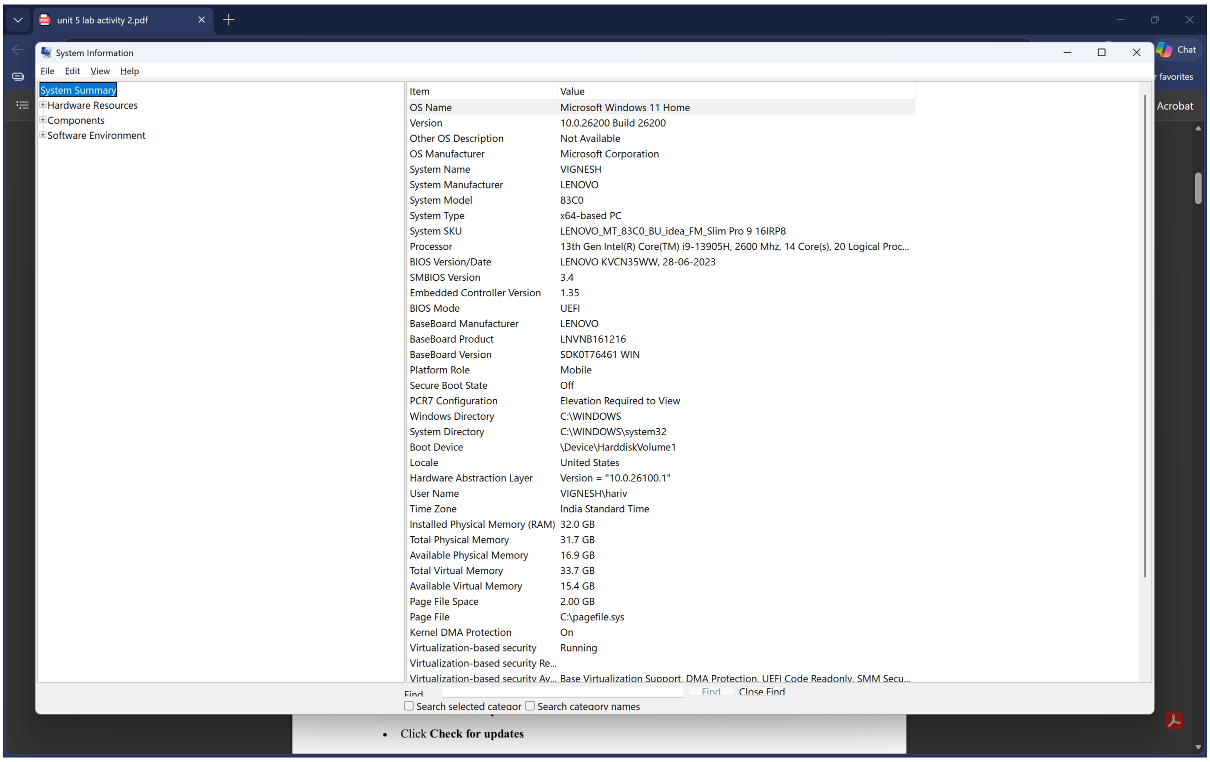
4. Procedure

STEP 1: System Information

- Press Windows + R
- Type: msinfo32

Note:

- OS Version
- System Type



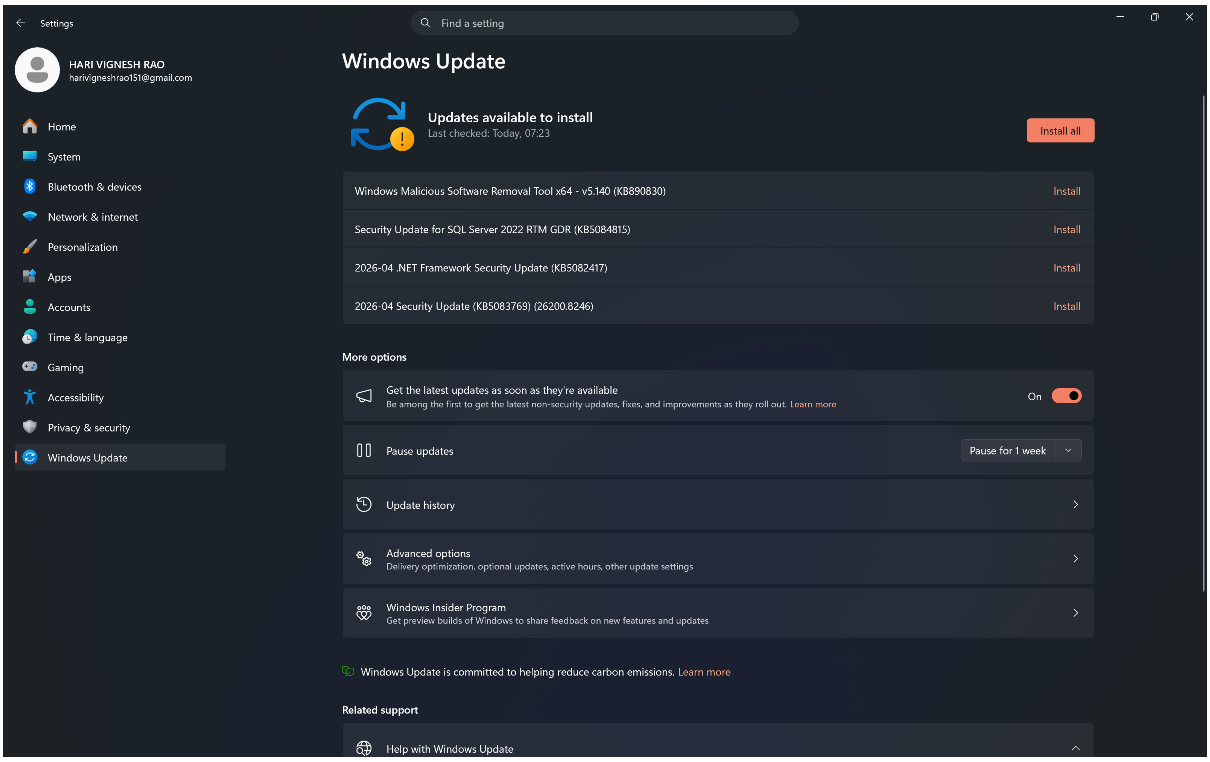
Observation:

The system information was obtained using the msinfo32 tool. The system is running on Microsoft Windows 11 Home Single Language, 64-bit operating system, with Intel i5 processor and UEFI based architecture. Secure Boot is enabled, indicating a secure boot configuration. Secure Boot being enabled enhances system security by preventing unauthorized boot loaders.

STEP 2: Check Windows Updates

Steps:

- Open Settings
- Go to Windows Update
- Click Check for updates



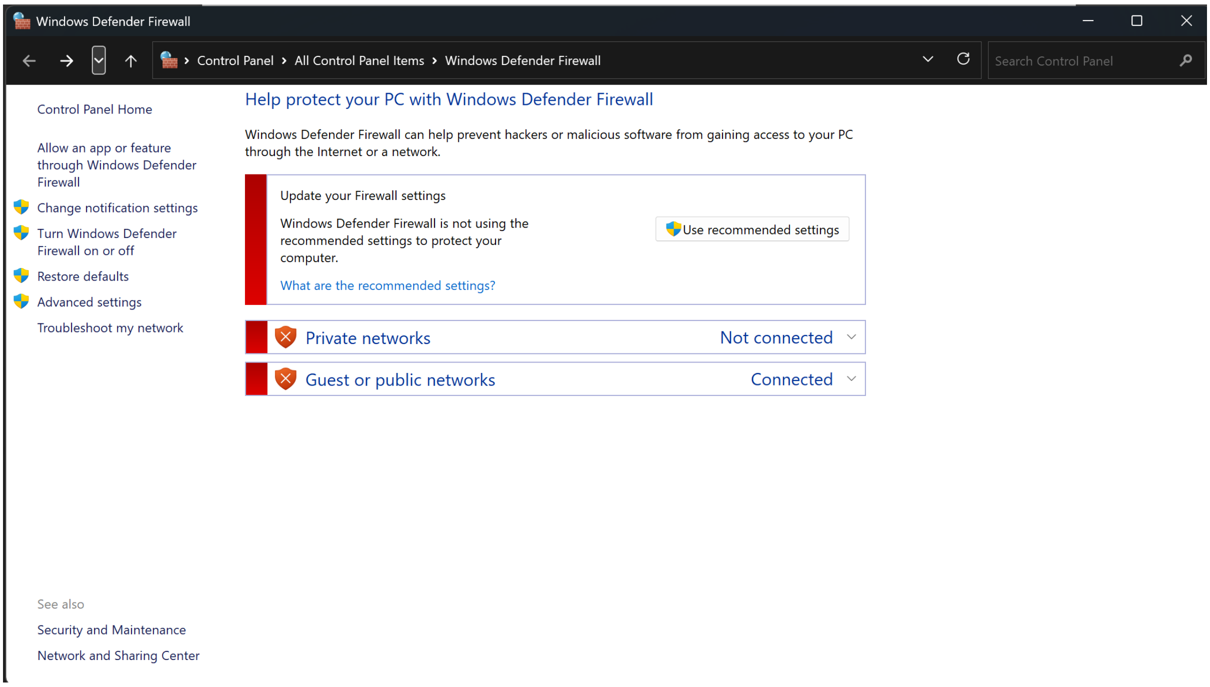
Observation:

The Windows Update section was checked, and it was found that updates are available and pending installation. The system requires a restart to complete the update process. Pending updates may expose the system to known vulnerabilities. Regular updates are essential to maintain system security and protect against threats.

STEP 3: Check Firewall Status

Steps:

- Search: Windows Defender Firewall
- Click Turn Windows Defender Firewall on or off



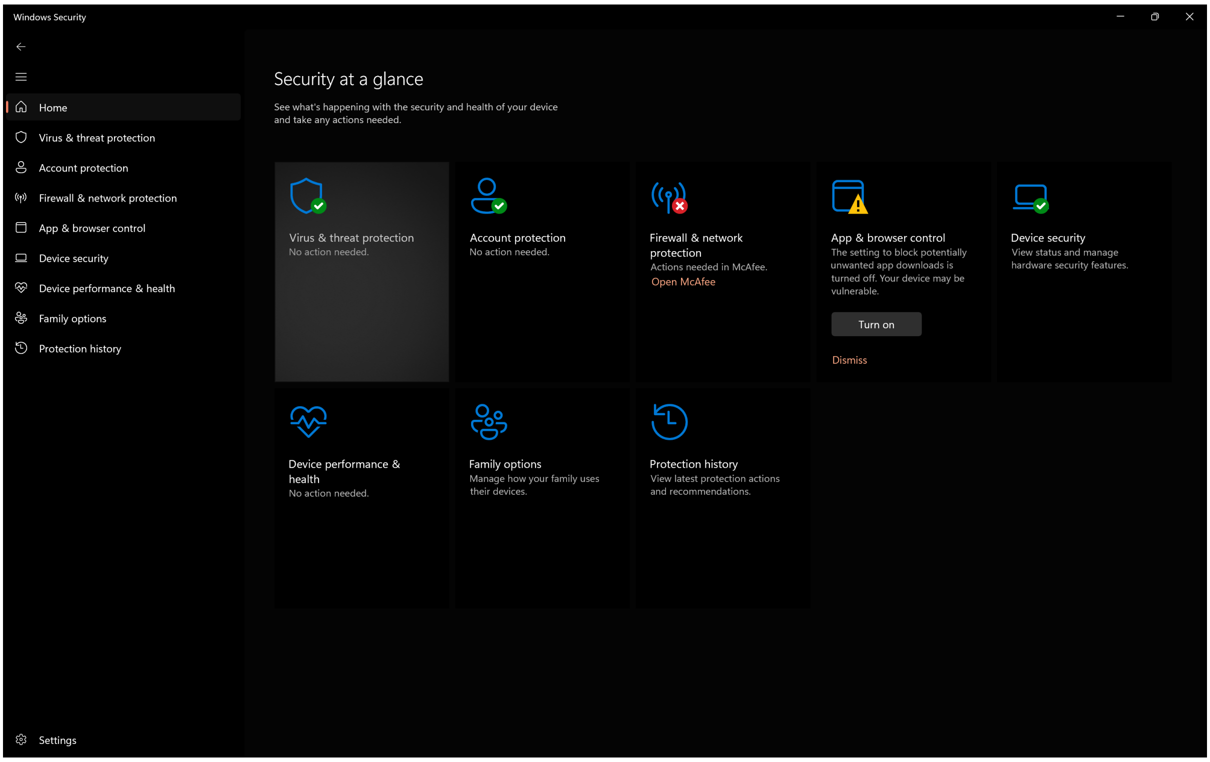
Observation:

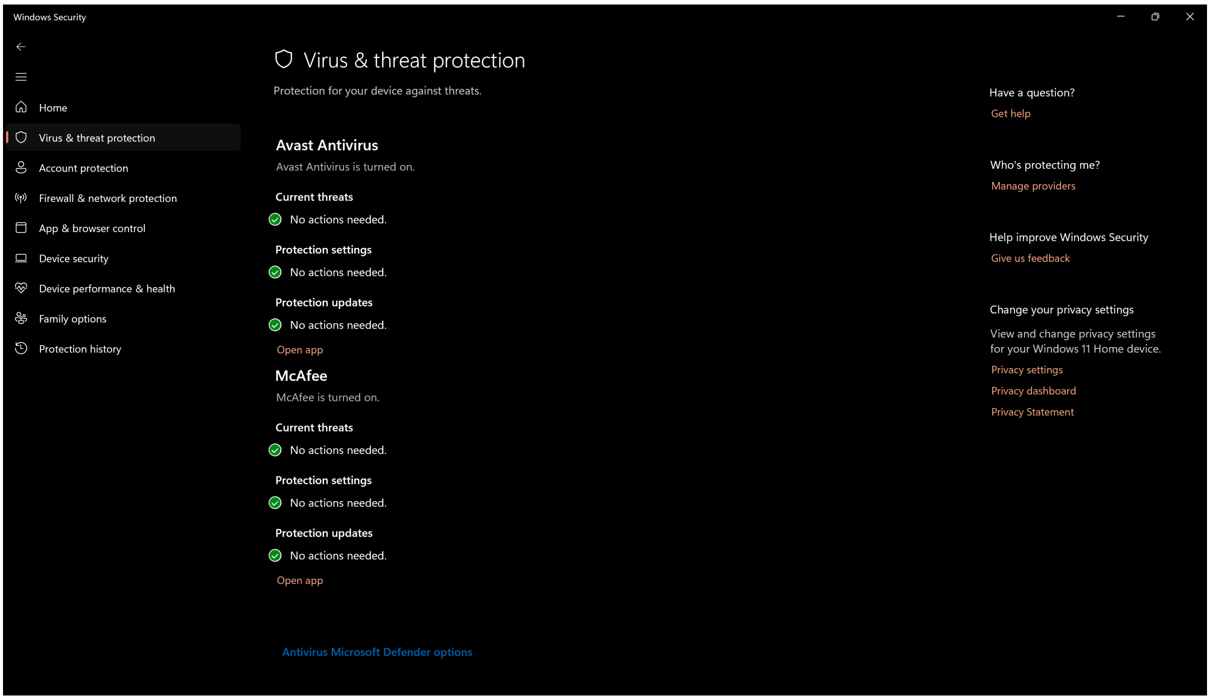
The firewall settings were checked using Windows Defender Firewall. It was observed that firewall settings are managed by Kaspersky antivirus, indicating that a third-party firewall is active on the system. The presence of a third-party firewall ensures network protection; however, misconfiguration may still expose the system to risks.

STEP 4: Check Antivirus (Windows Security)

Steps:

- Open Windows Security
- Go to Virus & Threat Protection Check:
- Real-time protection → ON
- Last scan → Recent





Observation:

The antivirus status was verified under Virus and Threat Protection. It was observed that Avast antivirus is active, and no current threats were detected. Protection settings and updates are functioning properly. A third-party antivirus like Avast provides advanced protection against malware, phishing, and other cyber threats.

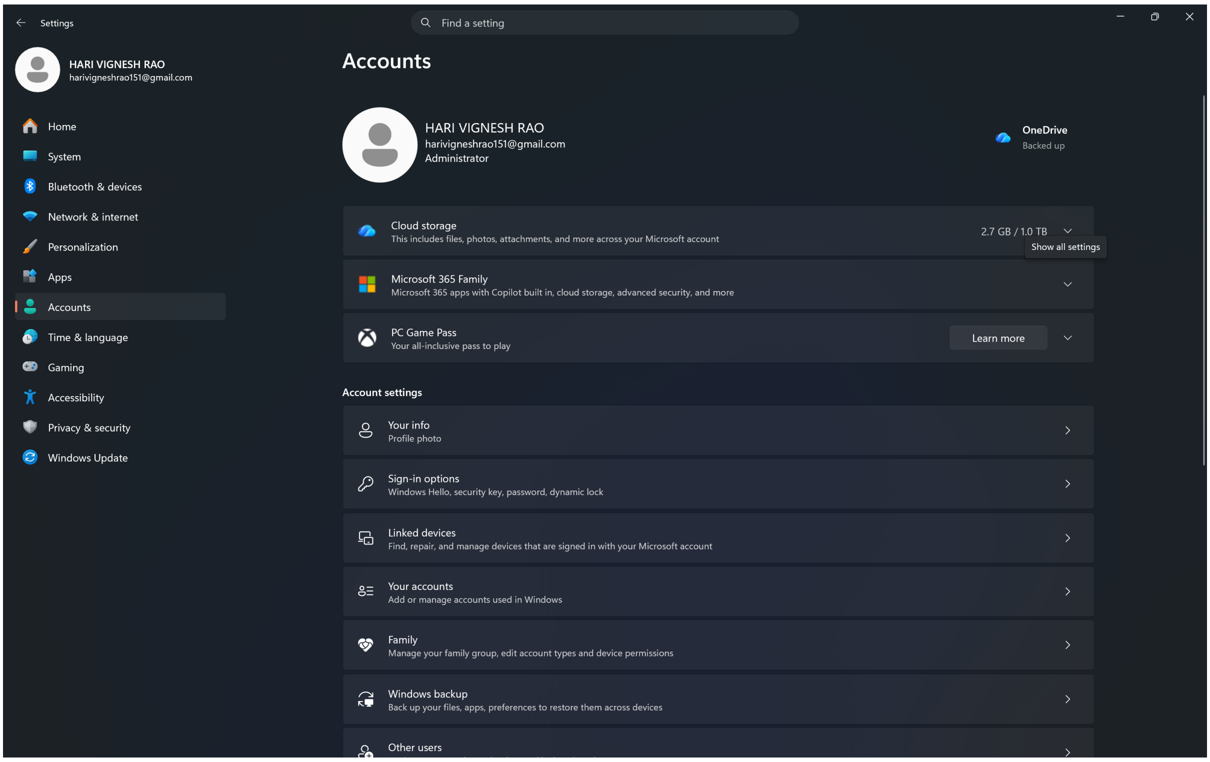
STEP 5: Check Password & Account Security

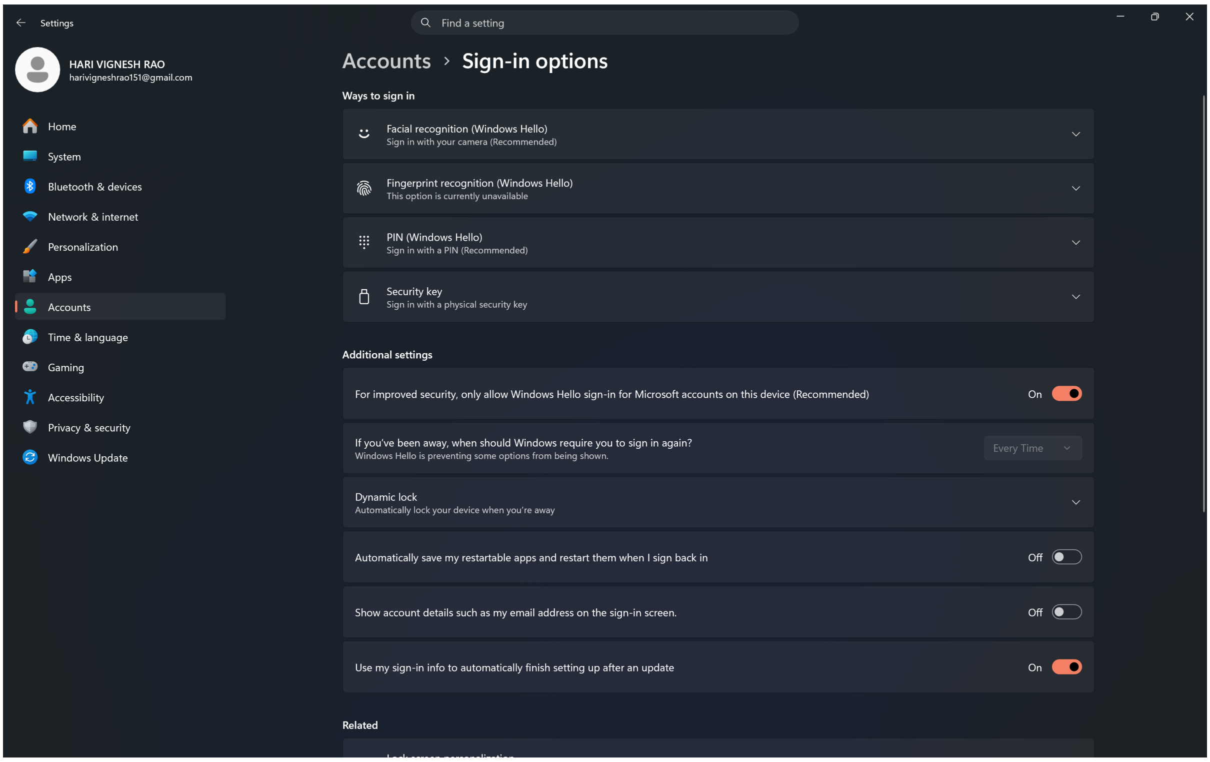
Steps:

- Settings → **Accounts → Sign-in options**

Check:

- Password / PIN enabled
- Windows Hello (optional)





Observation:

The account settings and sign-in options were checked. The system uses a local administrator account with multiple authentication methods such as password and PIN (Windows Hello), ensuring secure access.

Strong authentication mechanisms like PIN and biometric options enhance system security and prevent unauthorized access.

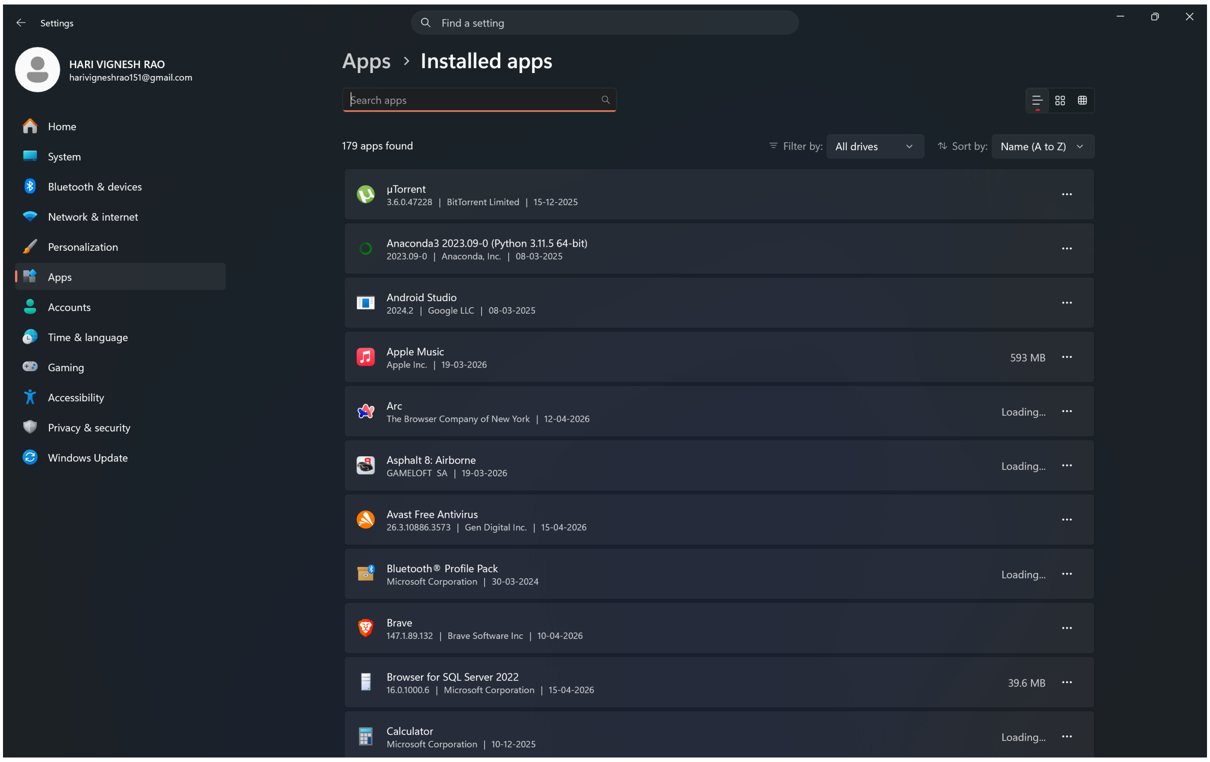
STEP 6: Check Installed Applications

Steps:

- Settings → **Apps → Installed apps**

Look for:

- Unknown software
- Cracked tools



Observation:

The list of installed applications was reviewed. No unauthorized or suspicious software was identified in the system. Unverified or cracked software may introduce malware and compromise system security.

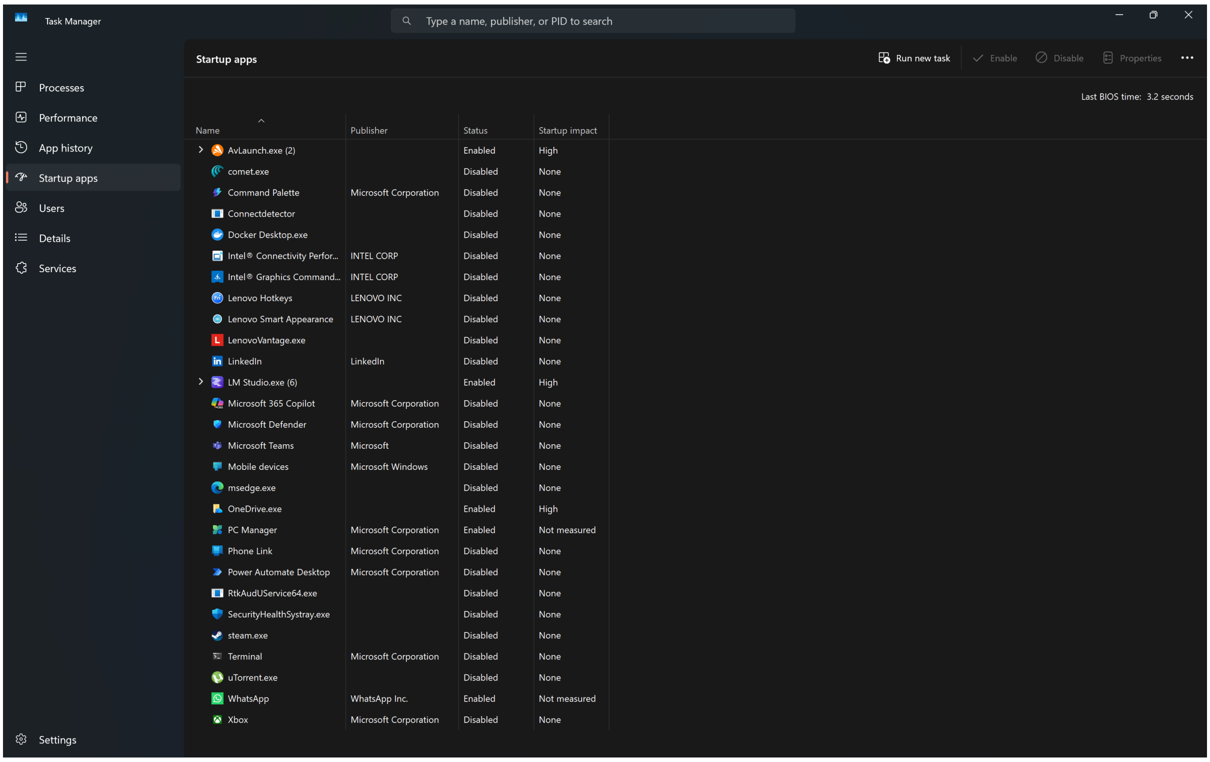
STEP 7: Check Startup Programs

Steps:

- Press **Ctrl + Shift + Esc**
- Go to **Startup tab**

Disable

: • Unknown apps



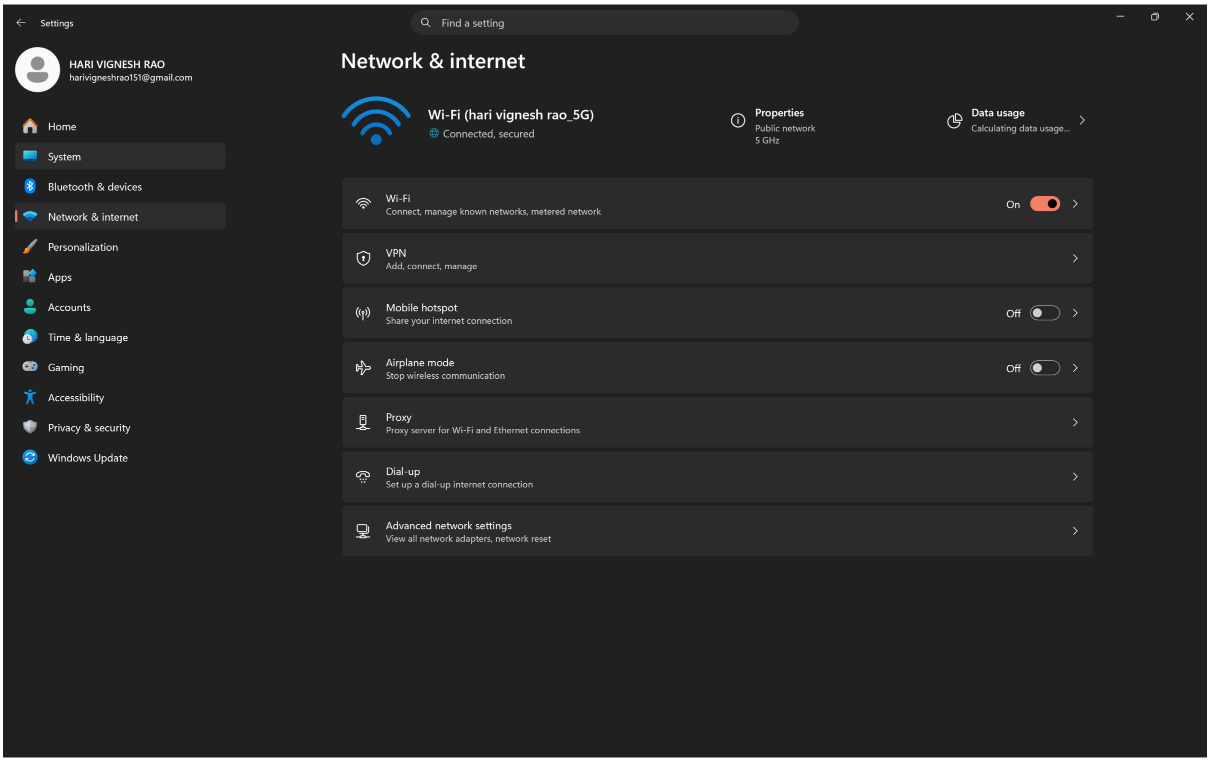
Observation:

Startup applications were analyzed using Task Manager. Some applications were enabled at startup, which may impact system performance and security. Disabling unnecessary startup applications reduces attack surface and improves system performance.

STEP 8: Check Network Security

Steps:

- Settings → **Network & Internet**
- Check:
    - o WiFi connection
    - o Network type → Private/Public



Network Overview

- Wi-Fi connected → **harivigneshrao_5G**
- Status → **Connected, secured**
- Shows general network info

Network Type

• Network profile → Public network (selected)

• Private option available

Observation:

The network settings were analyzed, and the system is connected to a Wi-Fi network with a public network profile. This restricts device visibility and enhances security.

Using a public network profile improves security by limiting unauthorized access, especially in untrusted environments.

STEP 9: Check Browser Security

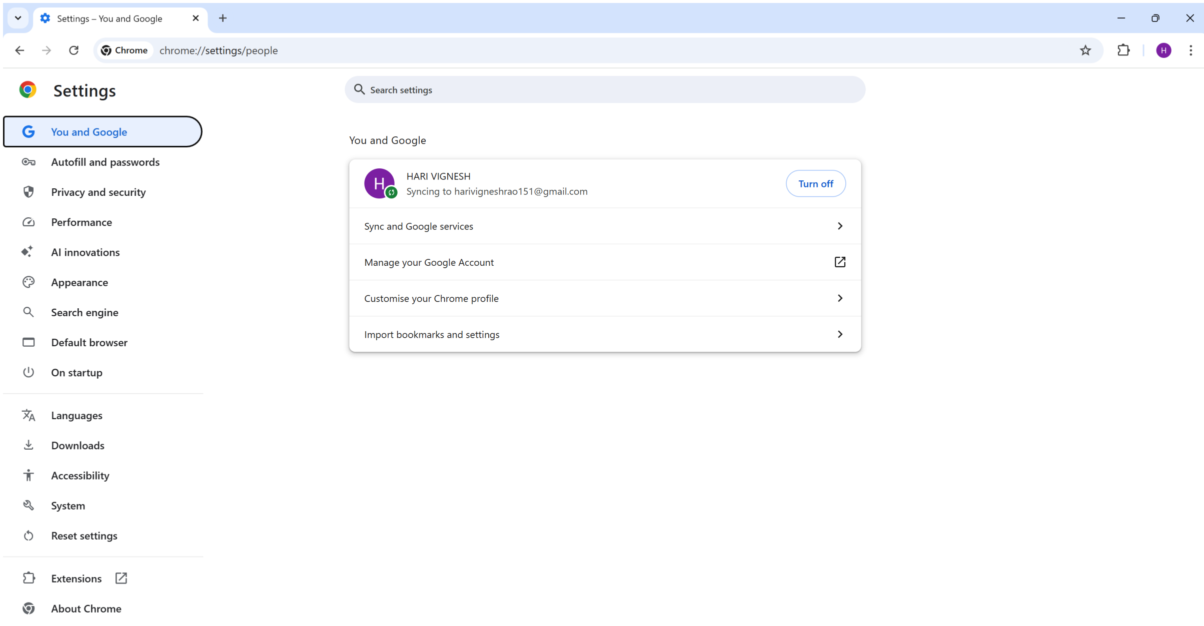
Steps (Chrome/Edge):

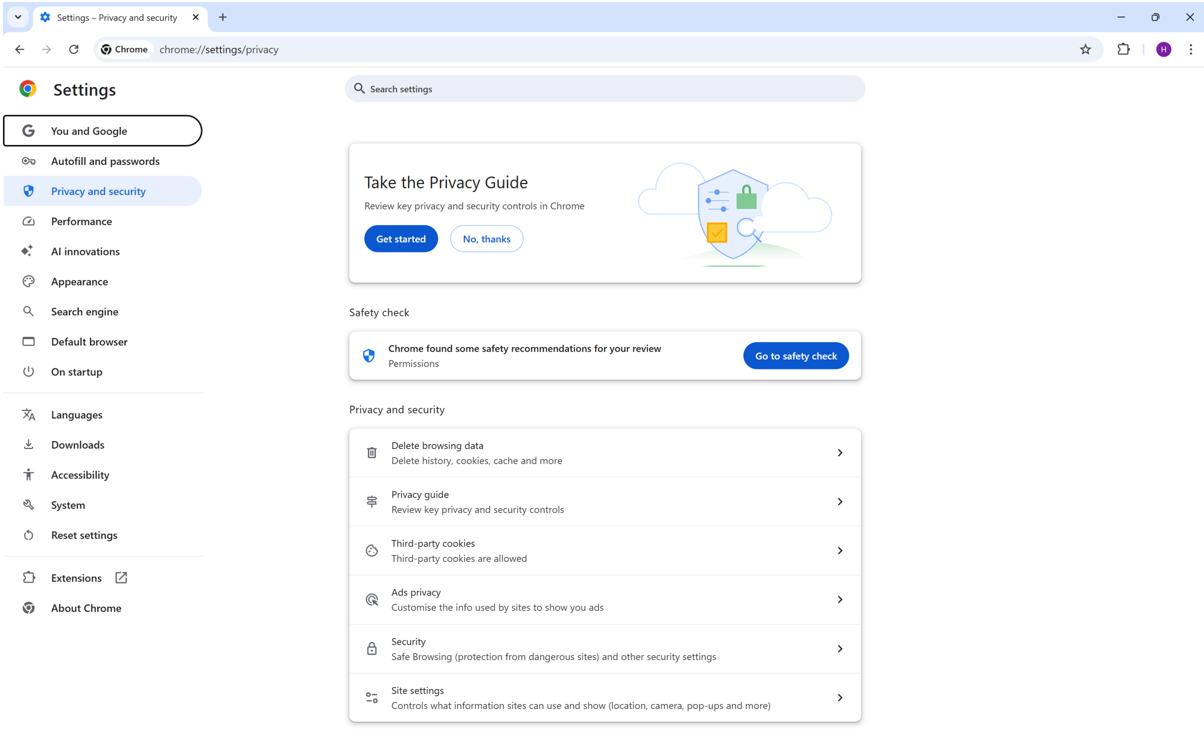
• Go to **Settings → Privacy & Security**

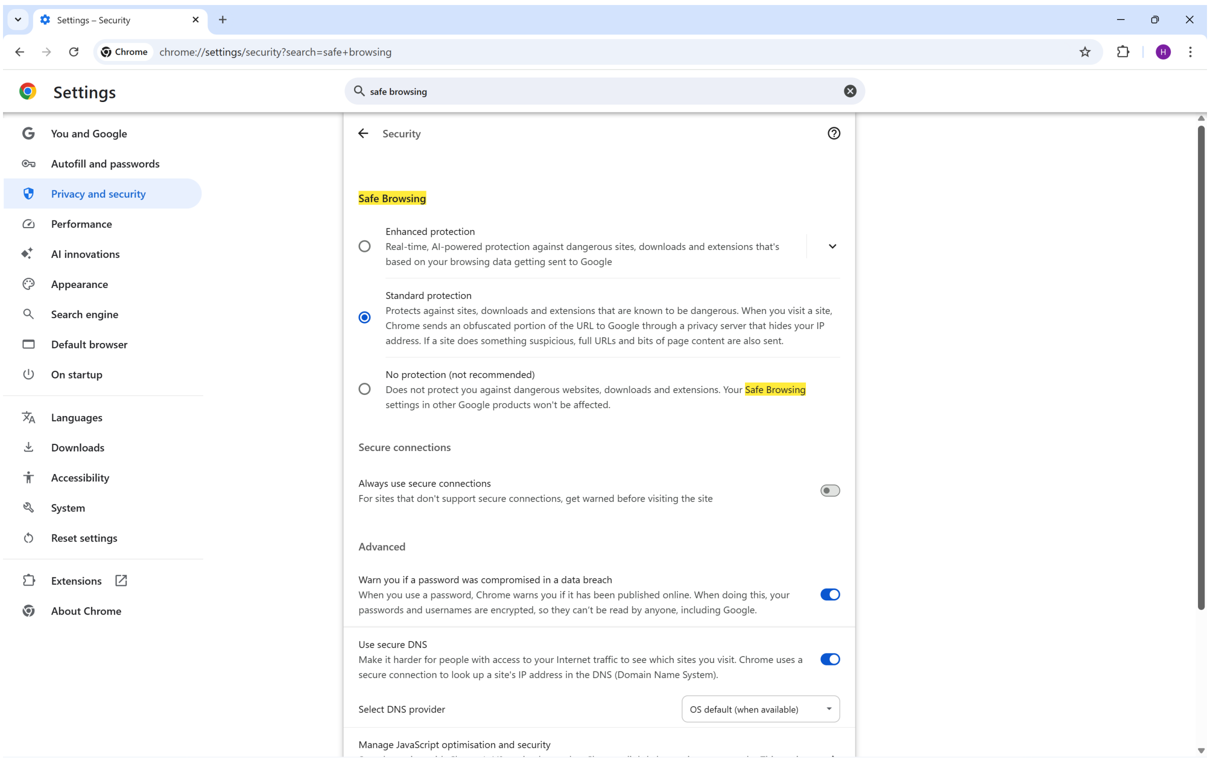
• Check:

  o Safe browsing ON

  o Remove unknown extensions







Observation:

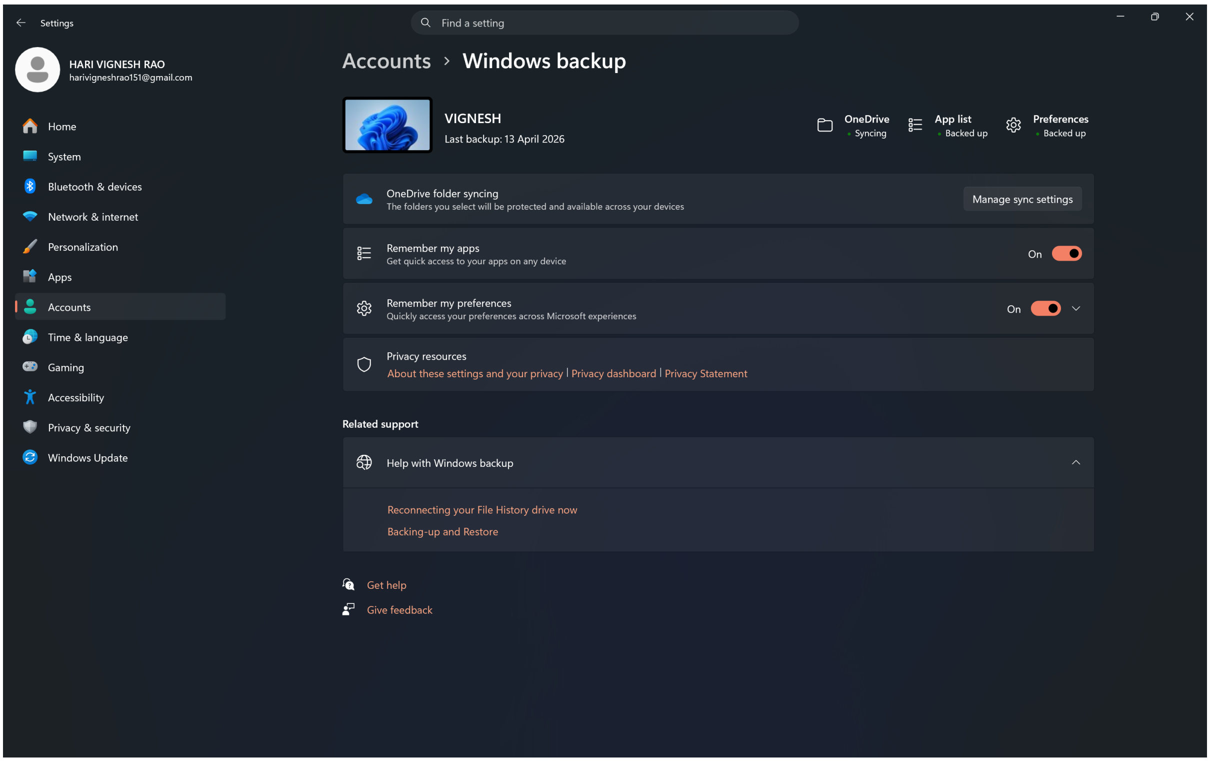
The browser security settings were examined under the Privacy and Security section. Safe Browsing was found to be enabled with standard protection, ensuring protection against malicious websites and unsafe downloads.

Safe Browsing helps prevent phishing attacks, malware infections, and unsafe browsing activities.

STEP 10: Check Data Backup

Steps:

- Search: **Backup settings**
- Check:
  - o OneDrive / External backup



Observation:

The backup settings were checked, and it was observed that OneDrive backup is not configured and system data is not being backed up.

Lack of proper backup mechanisms may lead to permanent data loss in case of system failure or cyber attacks such as ransomware.

STEP 11: Risk Assessment Table

Asset	Threat	Vulnerability	Risk Level	Solution
Personal Files	Data Loss	No Backup	High	Enable OneDrive
System	Malware	Unknown Apps	High	Remove apps
Network	Hacking	Public Network	Medium	Use Private
Account	Unauthorized Access	Weak Password	High	Strong password

STEP 12: Security Recommendations

✓ **Technical Controls**

- Enable firewall
- Enable antivirus
- Regular updates

✓ **Administrative Controls**

- Strong passwords
- Avoid unknown downloads

✓ **User Awareness**

- Avoid phishing links
- Safe browsing

STEP 13: Result

The system security audit identified potential vulnerabilities such as lack of backup and configuration risks. Suitable mitigation strategies were suggested to improve system security.